

MANAGEMENT REVIEW

Management without theatre.

STRATEGY

OPERATIONAL RISK: FMEA AND THE RISK MATRIX

Why the RPN takes only 120 values, two teams and one process, the Action Priority of 2019, what is wrong with risk matrices, and a card for failure modes.



120

different values are all the RPN can take, out of 1,000 combinations of severity, occurrence and detection.

BOWLES, 2003

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In this issue

Every operation keeps a list of what could go wrong. This issue asks how that list gets scored, why the most common scores mislead, and what to use instead to decide which risk to act on first.

FMEA began in the US military in 1949. Its Risk Priority Number multiplies three scores from 1 to 10, yet its 1,000 combinations give only 120 different values, and a severe failure can rank below a mild one. When two hospital teams analysed the same process, only 17 of their 50 failures each were the same. The 2019 AIAG & VDA handbook replaced the number with an Action Priority, and Tony Cox showed that risk matrices can rank smaller risks above larger ones.

Stiven Janaqi EDITOR · STIVEN CATALYST

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HOW TO READ THIS ISSUE

Figure

Every figure has its source and year at the foot of its page.

Our reading

Where the editors interpret rather than the research, it says so.

Practice

The steps and the card are proposals to try, not research results.

COVER STORY

120 out of 1,000

Failure Mode and Effects Analysis began in the US military, with procedure MIL-P-1629 of 1949. Its best-known number is the Risk Priority Number: severity $\times$ occurrence $\times$ detection, each scored from 1 to 10.

HOW THE 1,000 COMBINATIONS OF THE THREE SCORES SPREAD OVER THE RPN (EDITORS' CALCULATION)



The RPN looks like a scale from 1 to 1,000, but John Bowles showed that only 120 different values can occur: none lies between 901 and 999, and 120 itself comes from 24 different combinations. A severe failure can rank low: severity 10 with occurrence 1 and detection 1 gives 10, while $4 \times 4 \times 4$ gives 64.

OUR READING

Multiplying three ordinal scores produces a number that looks precise and is not. The three scores say more when read one by one.

Sources: U.S. Armed Forces, 1949; John B. Bowles, Annual Reliability and Maintainability Symposium (IEEE), 2003

The chart is the editors' calculation over all 1,000 combinations; the 120 values and the 24 ways to reach 120 follow Bowles.

THE NUMBERS

Two teams, one process

In two hospitals of the same NHS trust, two teams ran an FMEA of the same process at the same time: the use of the antibiotics vancomycin and gentamicin.

FAILURES FOUND BY EACH TEAM, AND BY BOTH



Their scores for severity and detectability, and their RPNs, differed markedly, so the failures were ranked differently. In a study of hazards in public leisure activities, David Ball and John Watt found that different assessors placed the same hazard very differently on a risk matrix, and the spread stayed large even after long reflection.

OUR READING

An FMEA says as much about the team as about the process. A second team, or a second look, finds what one team misses.

Sources: Nada Atef Shebl, Bryony Dean Franklin & Nick Barber, Journal of Patient Safety, 2009; David Ball & John Watt, Risk Analysis, 2013

17 of 50 is 34% for each team; the study's 17% is counted over all 100 failures. One process in one trust.

THE MODEL

From RPN to Action Priority

In June 2019 AIAG and VDA, the US and German automotive bodies, published a joint FMEA handbook. It sets out seven steps and replaces the RPN with an Action Priority, read from a table that gives severity the most weight, then occurrence, then detection.

Analysis

1 Planning and preparation

2 Structure analysis

3 Function analysis

4 Failure analysis

Risk and action

5 Risk analysis

6 Optimisation

7 Documenting the results (new)

01 High

Highest priority: improve prevention or detection, or document why the current controls are enough.

02 Medium

Find actions or, if the company decides so, document why the controls are enough.

03 Low

Actions may be found.

OUR READING

The question changes from “how big is the number?” to “what must we do?”. That is the question a team can act on.

Source: AIAG & VDA QMC, 2019

The grouping of the steps is the editors'. Some customers still accept the RPN alongside the Action Priority.

More in the essay

Why the shift handover is one of the most underrated processes

WHAT THE RESEARCH SAYS

What is wrong with risk matrices

In 2008 Tony Cox analysed risk matrices mathematically in the journal Risk Analysis. He found four kinds of problem.

01	Poor resolution	Typical matrices compare correctly only a small share of pairs of risks, in his example less than 10%, and give the same rating to very different risks.
02	Errors	They can rate smaller risks above larger ones; when frequency and severity are negatively related, they can be worse than useless.
03	Resources	Resources for reducing risk cannot be allocated well on the basis of their categories.
04	Judgment	Inputs and outputs need subjective interpretation; different users can rate the same risks in opposite ways.

In 2013 Philip Thomas, Reidar Bratvold and Eric Bickel found no published scientific study showing that risk matrices improve risk decisions. In one drilling example, reversing the scoring scale reversed the order of priorities.

OUR READING

A matrix is a picture of judgments, not a measurement. It can open a conversation; it should not close one.

Sources: Louis Anthony (Tony) Cox Jr., Risk Analysis, 2008; Philip Thomas, Reidar B. Bratvold & J. Eric Bickel, SPE Annual Technical Conference and Exhibition, 2013

Cox's results are mathematical, not measurements in organisations; the 10% is his example.

HOW IT IS MEASURED

Scoring risk without fooling yourself

ISO 31000 defines risk as the effect of uncertainty on objectives, positive or negative, and treats managing it as a process: identify, analyse, evaluate, treat, and keep monitoring and communicating.

1

List failures with the people who do the work

Two groups, or two sessions, find more than one.

2

Keep the three scores apart

Write down severity, occurrence and detection, not only their product.

3

Look at severity first

A failure with severity 9 or 10 gets an action or a written reason, whatever its RPN.

4

Score again after the action

If nothing was done, the risk has not changed.

HYPOTHETICAL EXAMPLE, TWO FAILURES IN A WAREHOUSE

Label Wrong label on a pallet: $4 \times 6 \times 3 = \text{RPN } 72$

Cold chain A medicine order left unrefrigerated: $10 \times 2 \times 3 = \text{RPN } 60$

The RPN puts the label first; severity puts the cold chain first. The failures and scores are invented.

Source: ISO, 2018

The steps and the example are the editors'; the scores are severity $\times$ occurrence $\times$ detection.

TOOL OF THE ISSUE

The failure-mode card

One card per process step. Fill it in with the people who do the work, and keep the three scores side by side.

01 Process step where in the process, and who does it

02 What can go wrong the failure mode, in the words of the people who see it

03 Effect on the customer, the next step or safety

04 Severity · occurrence · detection three scores from 1 to 10, written separately

05 Priority high, medium or low, and the reason

06 Action and check what we do, who, by when, and the scores afterwards

Source: AIAG & VDA QMC, 2019

A practice proposed by the editors, after the AIAG & VDA handbook (2019).

SOURCES AND METHOD

Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.

MIL-P-1629: Procedures for Performing a Failure Mode, Effects and Criticality Analysis
U.S. Armed Forces, 1949

An Assessment of RPN Prioritization in a Failure Modes Effects and Criticality Analysis
John B. Bowles, Annual Reliability and Maintainability Symposium (IEEE), 2003

Is failure mode and effect analysis reliable?
Nada Atef Shebl, Bryony Dean Franklin & Nick Barber, Journal of Patient Safety, 2009
<https://doi.org/10.1097/PTS.0b013e3181a6f040>

Further Thoughts on the Utility of Risk Matrices
David Ball & John Watt, Risk Analysis, 2013
<https://doi.org/10.1111/risa.12057>

AIAG & VDA FMEA Handbook: Design FMEA, Process FMEA, Supplemental FMEA for Monitoring & System Response
AIAG & VDA QMC, 2019
https://webshop.vda.de/QMC/en/aiag-vda-fmea-handbook_eng

What's Wrong with Risk Matrices?
Louis Anthony (Tony) Cox Jr., Risk Analysis, 2008
<https://doi.org/10.1111/j.1539-6924.2008.01030.x>

The Risk of Using Risk Matrices
Philip Thomas, Reidar B. Bratvold & J. Eric Bickel, SPE Annual Technical Conference and Exhibition, 2013
<https://doi.org/10.2118/166269-MS>

ISO 31000:2018 Risk management — Guidelines
ISO, 2018
<https://www.iso.org/standard/65694.html>

EDITORIAL METHOD

Each figure was checked for its year, its publisher and what exactly it measures. Where the publisher's page could not be opened, the figure was checked against independent summaries and is marked “via”. The editors' interpretation is marked “Our reading”. Figures that could not be confirmed are not in the issue.



MANAGEMENT REVIEW

Management without theatre.

Every issue, one management question, checked against the best research.

ALL ISSUES

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